

Infectious Discitis (Spondylodiscitis): A Comprehensive Disease-Characteristics Report

Disease: Infectious Discitis / Spondylodiscitis **Category:** Infectious Disease **Proposed MONDO:** the template-supplied MONDO:8000000 is a placeholder; the closest resolvable concepts are spondylitis/osteomyelitis of the spine (no distinct Mendelian entry) **Evidence base:** 14 confirmed findings, 65 papers reviewed across 5 investigation iterations

Summary

Infectious discitis — more precisely termed **spondylodiscitis** or **disco-vertebral infection** — is an acquired microbial infection of the intervertebral disc and the two adjacent vertebral endplates/bodies. It is fundamentally **not a genetic disease**: there are no causal genes, pathogenic germline variants, inheritance patterns, or heritable susceptibility loci. Consequently, all template sections concerning causal genes, ACMG variant classification, chromosomal abnormalities, inheritance, penetrance, carrier frequency, and genetic screening are **not applicable** and are marked as such below. The disease is driven overwhelmingly by **hematogenous seeding of *Staphylococcus aureus***, with a well-defined anatomical causal chain: bacteremia → seeding of the vascularized vertebral endplate → vertebral osteomyelitis → spread across the endplate into the avascular disc (discitis) → paravertebral/psoas and epidural abscess → spinal cord/root compression → neurologic deficit, deformity, sepsis, and death.

Epidemiologically the disease is a **rising-incidence condition of the aging population**. German population data show the age-adjusted incidence roughly doubled (5.4 → 11.0 per 100,000, +104%, $p < 0.001$) between 2005 and 2021, and Korean nationwide data show a 1.5-fold rise (22.90 → 35.79 per 100,000) from 2010–2019, with a simultaneous shift away from tuberculous toward pyogenic disease. The lumbar spine is most commonly affected (~51%), and the elderly (≥ 60 y) constitute ~48% of cases. A distinct, generally milder **pediatric form**

peaks at 0.5–4 years of age. Key risk factors are immunocompromise, bacteremic sources (endocarditis, IV drug use, hemodialysis), diabetes, and prior spinal instrumentation; malnutrition and substance-use disorder are emerging drivers.

Management rests on **contrast-enhanced whole-spine MRI plus microbiological confirmation** (blood cultures and image-guided disc/vertebral biopsy) supported by elevated CRP/ESR, and increasingly on **¹⁸F-FDG PET/CT** and **metagenomic next-generation sequencing (mNGS)** for inconclusive imaging and culture-negative disease (culture-negative rate ~43%). A landmark randomized trial established that **6 weeks of pathogen-directed antibiotics is non-inferior to 12 weeks**, with surgery reserved for instability, neurologic deficit, abscess, or medical failure. Prognosis is generally favorable (in-hospital mortality ~7%) but morbidity is substantial: 20–30% residual disability, 6–17% recurrence, and markedly worse outcomes with extensive epidural abscess, concurrent infection, thoracic involvement, and concomitant infective endocarditis.

Section 1 — Disease Information

Overview. Infectious discitis is an infection localized to the intervertebral disc that, in adults, almost always co-involves the adjacent vertebral endplates and bodies — hence the umbrella terms *spondylodiscitis*, *disco-vertebral infection*, *native vertebral osteomyelitis (NVO)*, and *pyogenic vertebral osteomyelitis (PVO)*. It is a devastating invasive infection that can lead to debilitating pain, motor weakness, or paralysis even with appropriate treatment ([PMID: 40429472](#)).

Key identifiers. - **ICD-10:** M46.4 (Discitis, unspecified); M46.3– (Infection of intervertebral disc, pyogenic); tuberculous spondylitis A18.01 / M49.0. - **ICD-11:** FA83 (Infective spondylopathies). - **MeSH:** *Discitis* (D053161); *Spondylitis*; *Osteomyelitis*. - **MONDO:** the supplied MONDO:8000000 is a placeholder; closest resolvable concepts are spondylitis/osteomyelitis of spine (not a distinct Mendelian entry). - **OMIM / Orphanet:** Not applicable — this is an acquired infection, not a Mendelian or rare genetic disorder, with no dedicated OMIM entry.

Synonyms: spondylodiscitis, disco-vertebral infection, discitis, infective/infectious discitis, pyogenic vertebral osteomyelitis, native vertebral osteomyelitis, septic discitis.

Information source: Aggregated disease-level resources — nationwide registries (German EANS Spine, Korean NHIS), systematic reviews/meta-analyses, and society guidelines (IDSA 2015, SPILF 2022) — supplemented by EHR-derived single-center cohorts.

Section 2 — Etiology

Primary cause: infectious. *Staphylococcus aureus* is the predominant organism in pyogenic disease. In a comparative series of 114 cases, pyogenic disease accounted for 78% (TB 16%, brucellar 6%), with *S. aureus* the causative agent in most pyogenic cases (PMID: 41882873). The pathogen spectrum **shifts with age and site**: in German nationwide 2024 data (10,801 cases), *Staphylococcus* spp. prevalence declined with increasing age (35.7% → 21.3%, $p < 0.001$) while *Enterococcus* (3.9% → 11.8%) and Enterobacterales (17.3% → 21.4%) rose, and lumbar sites carried more Enterobacterales/Enterococcus (PMID: 42778058). *S. aureus* bacteremia metastasizes to the spine as spondylodiscitis in ~20% of cases (PMID: 41905317).

Risk factors (environmental/acquired). Established factors include intravenous drug use, hemodialysis, prior spinal surgery, diabetes, chronic kidney/liver disease, prior invasive procedures, and bacteremia (PMID: 40429472; PMID: 42262500). Community-acquired and persistent *S. aureus* bacteremia increased metastatic infection risk **7.6-fold and 4.8-fold** respectively (PMID: 41905317). Substance-use disorder rose >3-fold within spinal-infection cohorts, paralleling a ~40% rise in spinal infections in Washington State (PMID: 30059485). **Malnutrition/hypoalbuminemia** independently predicted in-hospital mortality and 90-day readmission (PMID: 39773279).

Genetic risk / protective factors: Not applicable — no established causal variants, susceptibility loci, or protective alleles. **Protective factors** are environmental: control of bacteremic sources, glycemic control, nutritional optimization, and aseptic surgical technique.

Gene–environment interactions: Not applicable in a Mendelian sense; host immune competence (diabetes, immunosuppression, ESRD) modulates susceptibility to a purely exogenous infectious exposure.

Section 3 — Phenotypes

Phenotype	Type	Frequency	Suggested HPO
Back/spinal pain	Symptom	78–85%	HP:0003418 (Back pain)
Fever	Sign	29–76%	HP:0001945 (Fever)
Elevated CRP	Lab abnormality	~95%	HP:0011227 (Elevated CRP)
Bacteremia	Lab/sign	~76%	HP:0032243 (Bacteremia)
Elevated ESR	Lab abnormality	high (78% sens. in children)	HP:0003565 (Elevated ESR)
Motor weakness / paralysis	Sign	variable	HP:0001324 (Muscle weakness)
Refusal to walk/stand/sit (pediatric)	Behavioral/sign	49.8% (children)	gait abnormality
Altered mobility	Sign	~33%	HP:0002540

Typical adult presentation: back pain (78%), fever (76%), elevated CRP (95%), bacteremia (76%) ([PMID: 30002914](#)); in a Scottish cohort back pain 84.7%, altered mobility 33.3%, fever 29.2%, lumbar involvement 51.4% ([PMID: 35904356](#)). **Onset** is typically subacute/insidious; **severity** is variable; **progression** is progressive if untreated. **Quality of life** is impaired by chronic pain and, in a minority, permanent neurologic disability. Pediatric presentation is dominated by back pain (37.97%) and refusal to walk/stand/sit (49.79%) ([PMID: 33396379](#)).

Section 4 — Genetic/Molecular Information

Not applicable — acquired infectious disease. There are **no causal genes, pathogenic germline/somatic variants, modifier genes, disease-specific epigenetic signatures, or chromosomal abnormalities**. The relevant "molecular" biology is that of the **pathogen**, principally *S. aureus* virulence and biofilm regulation (see Section 6). In a USA300 MRSA implant-osteomyelitis model, the **msaABCR operon** regulated biofilm formation, capsule,

protease, persister cells, and cell-wall integrity; wild-type caused severe bone necrosis with sequestra and multinucleated osteoclasts, whereas the msaABCR mutant was attenuated (PMID: 33109085). Host genetic susceptibility loci are not established for this condition.

Section 5 — Environmental Information

Environmental/iatrogenic factors: spinal surgery and invasive procedures (direct inoculation), indwelling vascular catheters, hemodialysis access. **Lifestyle factors:** intravenous drug use and substance-use disorder (PMID: 30059485); poor nutritional status (PMID: 39773279).

Infectious agents (NCBI Taxonomy): - *Staphylococcus aureus* (txid1280) — predominant pyogenic agent. - **Enterobacterales** (e.g., *Escherichia coli* txid562) and *Enterococcus* spp. (txid1350) — increasing in the elderly and at lumbar sites (PMID: 42778058). - *Mycobacterium tuberculosis* (txid1773) — tuberculous spondylitis (Pott disease), declining in incidence but endemic regionally. - *Brucella* spp. (txid234) — brucellar spondylodiscitis in pastoral/endemic areas. - **Fungi** (*Aspergillus*, *Blastomyces*, *Candida*) — rare, in immunocompromised hosts. - **Anaerobes** (e.g., *Cutibacterium acnes*, vaginal anaerobes) — implicated in low-grade and culture-negative disease, often requiring molecular detection (PMID: 42611158).

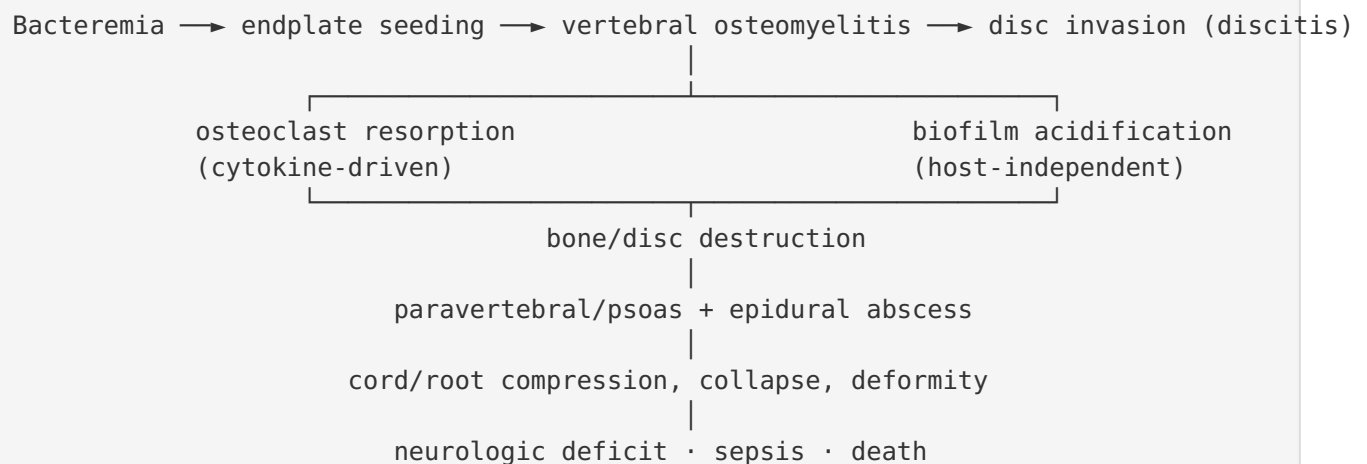
Section 6 — Mechanism / Pathophysiology

Ordered causal chain

1. **Bacteremia** (from skin, IV drug use, endocarditis, urinary, dental, or procedural source) **leads to** circulating pathogen reaching spinal vasculature.
2. Bloodborne organisms **seed the vascular vertebral endplate** (richly perfused via Batson's venous plexus / segmental arteries) — *hematogenous route is dominant* (PMID: 26700630).
3. Seeding **results in vertebral body osteomyelitis** with local proliferation.
4. Infection **spreads across the endplate into the avascular disc**, producing **discitis** (the disc has poor immune surveillance, favoring persistence).
5. Bacterial virulence and **biofilm formation lead to** bone destruction via **two branching mechanisms**: (a) host **osteoclast-mediated resorption** driven by inflammatory cytokines,

and (b) **direct biofilm acidification/dissolution** of bone independent of host immunity (PMID: 33109085; PMID: 28076372).

6. Local **inflammatory cytokines (IL-1 β , TNF- α , IL-6)** and **microthrombosis** amplify tissue injury and impair vertebral/disc blood supply (PMID: 18246002; PMID: 32775107).
7. Progression **results in paravertebral/psoas and epidural abscess** formation.
8. Abscess and bony destruction **lead to spinal cord/nerve-root compression, vertebral collapse and deformity**.
9. Endpoint: **neurologic deficit, sepsis, and death** — "any delay in diagnosis and treatment can lead to significant spinal cord injury, permanent neurological damage, septicemia, and death" (PMID: 35706345). Epidural abscess extending ≥ 3 segments raised lethality (HR 4.72, 95%CI 1.57–14.20, $p=0.006$) (PMID: 35657387).



Molecular/cellular processes: NF- κ B-driven inflammation, cytokine cascades (IL-1 β , TNF- α , IL-6), osteoclastogenesis (RANKL axis inferred), neutrophil recruitment, microthrombosis. **Upstream** events are bacteremia and endplate seeding; **downstream** are abscess, compression, and systemic sepsis.

Suggested ontology terms: GO:0006954 (inflammatory response), GO:0045124 (regulation of bone resorption), GO:0042116 (macrophage activation), GO:0006935 (chemotaxis); CL:0000092 (osteoclast), CL:0000062 (osteoblast), CL:0000775 (neutrophil); CHEBI:16412 (lipopolysaccharide).

Section 7 — Anatomical Structures Affected

- **Primary structures:** intervertebral disc (UBERON:0006678) and adjacent **vertebral endplates/bodies** (UBERON:0002412 vertebral column). **Lumbar spine most commonly affected (~51%)** (PMID: 35904356); L7–S1 in canine analogues.
 - **Secondary involvement:** epidural space (epidural abscess), psoas/paravertebral muscles (psoas abscess), spinal cord (UBERON:0002240) and nerve roots (compression), bloodstream (sepsis), and heart valves (concurrent infective endocarditis).
 - **Body systems:** musculoskeletal (primary), nervous (secondary compression), cardiovascular (bacteremia/endocarditis source).
 - **Tissue/cell level:** cartilaginous/fibrocartilaginous disc (nucleus pulposus, annulus fibrosus), trabecular bone, endothelium. Cell types: osteoclasts (CL:0000092), osteoblasts (CL:0000062), neutrophils (CL:0000775), nucleus pulposus cells.
 - **Subcellular:** bacterial intracellular persistence and biofilm on bone/implant surfaces; cytoplasmic inflammasome activation (GO:0005737 cytoplasm).
 - **Localization/lateralization:** typically midline/bilateral involving two contiguous vertebrae and the interposed disc; abscesses may be asymmetric (unilateral psoas).
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Section 8 — Temporal Development

- **Onset:** subacute to insidious in adults; median symptom duration before diagnosis is often weeks. A distinct **pediatric form peaks at 0.5–4 years** (PMID: 33396379).
 - **Progression:** progressive if untreated — from discitis to abscess and neurologic compromise. Bimodal age distribution (pediatric vs adult/elderly).
 - **Course/duration:** the **pediatric form is frequently benign/self-limiting** but risks deformity/neurologic complications if diagnosis is delayed (PMID: 36000507); the **adult/elderly form predominates and carries higher mortality** (PMID: 37384614).
 - **Critical intervention window:** early antibiotics and, when indicated, **timely surgery (≤ 14 days)** significantly improve survival (see Section 11).
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Section 9 — Inheritance and Population (Epidemiology)

Epidemiology (rising incidence). German population-based data (2005–2021): age-adjusted incidence rose **104%, from 5.4 to 11.0 per 100,000 ($p < 0.001$)**, with inpatient mortality up 347% ([PMID: 37980371](#)). Korean nationwide data (2010–2019): infectious spondylodiscitis rose **1.5-fold from 22.90 to 35.79 per 100,000**; pyogenic rose 15.35 → 33.75 while tuberculous fell 7.55 → 2.04 per 100,000; elderly $\geq 60y$ = **47.6%** of cases; mean age 58y ([PMID: 37384614](#)).

Metric	Value	Source
German incidence 2005 → 2021	5.4 → 11.0 / 100,000 (+104%)	PMID: 37980371
Korean incidence 2010 → 2019	22.90 → 35.79 / 100,000 (1.5×)	PMID: 37384614
Elderly ($\geq 60y$) share	47.6%	PMID: 37384614
Lumbar involvement	~51%	PMID: 35904356
Pediatric peak age	0.5–4 years	PMID: 33396379

Sex ratio: male predominance in adults (and in canine analogues); pediatric series show near-equal (F:M 1.3:1) ([PMID: 31848723](#)). **Geographic variation:** tuberculous and brucellar forms concentrate in endemic/pastoral regions; pyogenic disease predominates in high-income aging populations.

Inheritance, penetrance, expressivity, anticipation, mosaicism, founder effects, consanguinity, carrier frequency: All Not Applicable — acquired infectious disease with no genetic transmission.

Section 10 — Diagnostics

Imaging. Contrast-enhanced MRI is the modality of choice (87–100% of specialists) and should cover the **entire spine in ≥ 2 orthogonal planes** ([PMID: 36894008](#); [PMID: 36690329](#)). **^{18}F -FDG PET/CT** is a highly sensitive adjunct — meta-analytic sensitivity ~95%, specificity 88–91% — conditionally recommended when MRI is inconclusive, contraindicated, or artifact-limited, and to define healed status ([PMID: 42551483](#); [PMID: 42533326](#)). PET/CT also helps differentiate tuberculous from pyogenic disease ([PMID: 42650944](#)).

Laboratory/biomarkers. Elevated **CRP** (~95%) and **ESR**; in children ESR is more sensitive than CRP (78% vs 38%) (PMID: 31848723). Leukocytosis is more prominent in pyogenic than brucellar disease. Suggested LOINC: CRP (1988-5), ESR (4537-7).

Microbiology. **Blood cultures before therapy** (positive ~70–76%); if negative, obtain **multiple image-guided disc/vertebral biopsy samples** (PMID: 36690329). **Culture-negative disease is common — pooled prevalence 43.2% (95%CI 33.8–52.6)** (PMID: 42262500). **Metagenomic NGS markedly improves pathogen detection — pooled sensitivity 89.7% (95%CI 85.6–93.1%)** versus culture (PMID: 41584315); disc culture out-yields vertebral bone (50.0% vs 27.8%, p=0.046) and mNGS augments detection further (PMID: 41407999). mNGS/16S resolve fastidious anaerobes and uncultivable organisms (PMID: 42611158; PMID: 41841243).

Clinical criteria & differential diagnosis. IDSA 2015 (PMID: 26229122) and SPILF 2022 (PMID: 36690329) provide evidence-based diagnostic/treatment frameworks. **Differential diagnosis:** tuberculous spondylitis (longer symptom duration, night sweats, thoracic involvement, abscess, vertebral collapse), brucellar spondylitis (lumbar, less destructive), and degenerative **Modic type-1 changes** (PMID: 33198439; PMID: 17624684). Fever (72.7% vs 45.2%, p=0.004) and sweating (72.7% vs 47.9%, p=0.009) favor brucellar over pyogenic in one comparison (PMID: 33198439).

Genetic testing / newborn screening / karyotyping / repeat expansion / CMA: Not applicable.

Section 11 — Outcome / Prognosis

Mortality. In-hospital mortality 7.1% across 10,801 cases (PMID: 42778058); 11–15% in higher-acuity cohorts (PMID: 35657387; PMID: 30002914). **Concurrent infection raised mortality ~4-fold (3.3% vs 0.8%)** (PMID: 37148755).

Morbidity. Residual disability in 20–30%, recurrence 6–17% (PMID: 42262500). Neurologic recovery occurred in only 27% of spinal-cord-injured patients (PMID: 35657387).

Prognostic factors. - **Epidural abscess extension ≥ 3 segments:** HR 4.72 (95%CI 1.57–14.20, p=0.006) for lethality (PMID: 35657387). - **Surgical timing:** timely surgery ≤ 14 days improved 1-year OS (HR 0.22, p=.019) and recurrence-free survival (HR 0.35, p=.024); prolonged time-to-culture-positivity ≥ 60 h (lower bacterial load) improved OS (HR 0.18, p=.003) (PMID: 42314843). - **Concurrent infective endocarditis (IE):** co-occurs in ~8% of IE patients (PMID: 33858337); in combined IE+SD requiring surgery, operating on SD **first** independently predicted 30-day

mortality (25.7% vs 11.4%, $p=.037$) (PMID: 38964339). - Age, thoracic involvement, limb weakness, urinary symptoms/cauda equina, and malnutrition worsen outcomes (PMID: 35904356; PMID: 39773279).

Section 12 — Treatment

Pharmacotherapy (mainstay). Prolonged **pathogen-directed antibiotics**. The landmark **Bernard et al. open-label non-inferiority RCT** (NCT00764114; 359 adults, 71 French centers) established that **6 weeks is non-inferior to 12 weeks** for pyogenic vertebral osteomyelitis (non-inferiority margin 10%) (PMID: 25468170). **SPILF 2022** recommends short, adapted courses with **early IV-to-oral switch**, reserving >6 weeks for complicated disease (PMID: 36690329). Real-world durations run longer (median ~92 days; German survey median total 8 weeks, IV 2 weeks) (PMID: 30002914; PMID: 36894008). Empiric regimens target *S. aureus* (including MRSA where prevalent) pending cultures. **NCIT: antibiotic therapy** (NCIT:C15219 Antibiotic Therapy).

Surgical/interventional. Most cases (82–86%) are managed conservatively; surgery in **14–40%** for instability, abscess, neurologic deficit, or medical failure (PMID: 37384614; PMID: 42262500; PMID: 26700630). Surgical toolkit: - **Anterior debridement + stabilization + fusion with titanium mesh cage** — systematic review of 192 patients; at admission 48% had abscess and 51% neurologic impairment; gram-positive 71%, gram-negative 24% (PMID: 27324195). - **Transforaminal lumbar interbody debridement and fusion (TLIDF)** for postdiscectomy discitis — 100% infection clearance, superior alignment/function vs conservative care (PMID: 30308339). - **Minimally invasive percutaneous disc irrigation/debridement** (dual Jamshidi needles) — feasible/safe for refractory disease in comorbid elderly (mean 67y) (PMID: 30123957). - **Local antibiotic carriers** (calcium sulfate/hydroxyapatite pellets) may shorten systemic therapy (PMID: 32719918).

Supportive/rehabilitative: pain control, immobilization/bracing, nutritional support, physiotherapy. **Sequence matters** in combined IE+SD — treat IE surgically first to reduce 30-day mortality (PMID: 38964339).

Gene/cell/RNA/targeted/immunotherapy, pharmacogenomics: Not applicable.

Section 13 — Prevention

- **Primary prevention:** control of bacteremic sources — aseptic vascular access and hemodialysis technique, harm-reduction for IV drug use, prompt treatment of *S. aureus* bacteremia, and infection-control during spinal surgery.
 - **Secondary prevention:** early recognition (high index of suspicion for back pain + fever + elevated CRP), and **cardiac screening (echocardiography)** given ~8% IE co-occurrence (PMID: 33858337).
 - **Tertiary prevention:** guideline-concordant antibiotic duration, timely surgery for abscess/deficit, and management of malnutrition to reduce mortality/readmission (PMID: 39773279).
 - **Immunization / genetic screening / carrier screening:** Not applicable (no vaccine for pyogenic disease; BCG relevant only to TB).
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Section 14 — Other Species / Natural Disease

Naturally occurring discospondylitis in dogs closely mirrors human hematogenous spondylodiscitis, providing a spontaneous large-animal comparative model. In a multi-institutional study of 386 dogs, males were overrepresented (236/386), L7–S1 was the most common site, and *Staphylococcus* species were prevalent (23/38 positive blood cultures); trauma increased relapse risk (OR 9.0, 95%CI 2.2–37.0), as did prior steroid therapy (PMID: 37288966). Mycotic (fungal) discospondylitis in dogs is predominantly *Aspergillus* (7/11), German Shepherd-predisposed, with poor prognosis (median survival 30 days) (PMID: 40270003). Disseminated fungal infections (e.g., *Rasamsonia*) cause multifocal discospondylitis in German Shepherds via urinary/hematogenous routes (PMID: 34387899).

- **Taxonomy:** *Canis lupus familiaris* (NCBI txid9615); *S. aureus* (txid1280).
 - **Breed (VBO):** German Shepherd predisposition, especially for mycotic forms.
 - **Zoonotic potential:** the disease itself is not zoonotic, but shared *S. aureus* biology underlies cross-species pathogenesis. Male predominance, lumbosacral predilection, and staphylococcal dominance are conserved across humans and dogs, supporting evolutionary conservation of the hematogenous-seeding mechanism.
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Section 15 — Model Organisms

- **Spontaneous large-animal model:** canine discospondylitis (above) — high face validity for hematogenous, *Staphylococcus*-driven, lumbosacral disease (PMID: 37288966).
 - **Induced rodent models:** rat **disc-stab injury models** demonstrate cytokine (IL-1 β , TNF- α , IL-6/IL-8) production and MAPK (p38/ERK/JNK) signaling underlying disc inflammation and degeneration (PMID: 18246002); useful for the inflammatory/tissue-damage arm rather than infection per se.
 - **Murine implant-associated osteomyelitis:** USA300 MRSA K-wire model dissects *S. aureus* biofilm/virulence (msaABCR operon) and osteoclast-mediated bone destruction (PMID: 33109085).
 - **In vitro/ex vivo biofilm-on-bone systems:** demonstrate direct, host-independent bone dissolution by *S. aureus*, *P. aeruginosa*, *C. albicans*, *S. mutans* biofilms via acidification (PMID: 28076372).
 - **Limitations:** rodent disc-injury models capture inflammation but not spontaneous hematogenous seeding; implant models emphasize device-associated rather than native disc infection.
 - **Resources:** MGI (mouse), RGD (rat), plus veterinary case registries (VetCompass) for canine natural disease.
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Mechanistic Model / Interpretation

The unifying model is **hematogenous metastasis of a bacteremic focus to the richly vascularized vertebral endplate, followed by centrifugal spread into the immunologically privileged, avascular disc**. Two destructive engines then operate in parallel — **host osteoclast resorption** (cytokine/RANKL-driven) and **pathogen biofilm acidification** (host-independent). Biofilm additionally explains chronicity, culture-negativity (~43%), and implant-associated recurrence, and rationalizes the need for both prolonged antibiotics and, when biofilm-laden necrotic tissue/hardware is present, surgical debridement. The clinical severity gradient — from self-limiting pediatric discitis to lethal elderly spondylodiscitis with epidural abscess — is governed by host factors (age, immunocompetence, nutrition), bacterial burden (time-to-culture-positivity as a surrogate), and abscess extent. Prognostic levers that are actionable — **early antibiotics, timely surgery ≤ 14 days, correct IE-first sequencing, and nutritional optimization** — all map cleanly onto this causal chain.

Evidence Base (selected)

PMID	Contribution	Role
37980371	+104% incidence rise (Germany)	Supports rising epidemiology
37384614	1.5× rise, TB decline, elderly 47.6% (Korea)	Supports epidemiology/age
42778058	Age/site pathogen shift; 7.1% mortality (n=10,801)	Supports etiology & prognosis
41905317	<i>S. aureus</i> bacteremia → spine 20%; 7.6×/4.8× risk	Supports pathogenesis/risk
42262500	Culture-negative 43.2%; recurrence/disability	Supports diagnostics/prognosis
26700630 / 35706345	Three routes; downstream complications	Supports causal chain
35657387	Abscess ≥3 segments HR 4.72; 27% neuro recovery	Supports prognosis
36690329 / 26229122	SPILF 2022 / IDSA 2015 guidelines	Supports diagnosis/treatment
25468170	6 vs 12 wk RCT non-inferiority	Landmark treatment evidence
42551483 / 42533326	FDG-PET/CT ~95% sens; adjunct recommendation	Supports diagnostics
41584315 / 41407999	mNGS 89.7% sens; disc>bone yield	Supports molecular diagnostics
33109085 / 28076372	Biofilm/osteoclast bone destruction	Supports mechanism
37288966 / 40270003	Canine discospondylitis model	Supports comparative biology
42314843 / 38964339 / 33858337	Surgical timing; IE co-occurrence/sequencing	Supports prognosis

Limitations and Knowledge Gaps

1. **Observational evidence dominates.** Apart from the Bernard 6-vs-12-week RCT, most treatment and prognostic data are retrospective; surgical indications and technique choice remain heterogeneous ([PMID: 36894008](#)).
 2. **Culture-negativity (~43%)** limits pathogen-directed therapy; mNGS performance, while strong (89.7% sensitivity), still needs standardization, cost-effectiveness data, and contamination controls.
 3. **No host-genetic susceptibility data** — whether polymorphisms in innate immunity modulate risk is unstudied; the disease is treated here as purely acquired.
 4. **Prognostic surrogates** such as time-to-culture-positivity are promising but derive from small cohorts (n=48) ([PMID: 42314843](#)).
 5. **Pediatric vs adult forms** may represent partly distinct entities; the self-limiting pediatric form is under-characterized microbiologically.
 6. **Low-grade disc infection** (e.g., *Cutibacterium acnes*) and its relationship to Modic changes/low back pain remains controversial and outside classic spondylodiscitis.
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Proposed Follow-up Experiments / Actions

1. **Prospective mNGS validation trial** in culture-negative spondylodiscitis, comparing time-to-effective-therapy and outcomes vs culture-guided care.
2. **Randomized evaluation of early surgery (≤ 14 days)** vs antibiotics-first in patients with epidural abscess but no deficit, to test the observed survival benefit prospectively.
3. **Biomarker study** validating time-to-culture-positivity and CRP/ESR trajectories as prognostic tools in a multicenter cohort.
4. **Anti-biofilm adjunct trials** (e.g., rifampin combinations, local antibiotic carriers) building on [PMID: 32719918](#).
5. **Standardized IE screening protocol** (routine echocardiography) evaluation given ~8% co-occurrence and the mortality impact of surgical sequencing.
6. **Comparative canine–human genomic/microbiome studies** leveraging the spontaneous dog model to dissect host-independent biofilm bone destruction.

7. Nutritional intervention trial targeting hypoalbuminemia to test whether correcting malnutrition improves mortality/readmission.

Report compiled from 14 confirmed findings and 65 reviewed papers over 5 investigation iterations. Sections on causal genes, inheritance, variant classification, chromosomal abnormalities, pharmacogenomics, gene/cell/RNA therapy, and genetic screening are marked Not Applicable because infectious discitis is an acquired microbial infection, not a heritable disorder.

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